OOI KAI HONG

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PROFESSIONAL PROFILE

Interdisciplinary biomedical researcher with experience in cell biology, advanced microscopy, quantitative bioimage analysis, clinical research, trial documentation, research ethics and scientific writing. Developed OrganelleScope, an AI-assisted desktop microscopy workbench combining earlier colocalization and line-profiling tools, and contributed to an object-detection workflow for microtubule analysis. Currently pursuing a Master of Artificial Intelligence at Monash University, with developing Python and AI/ML skills supported by structured study and practical projects.

CORE COMPETENCIES

Biomedical Research	Cell Biology | Molecular Biology | Cell Culture | Biochemistry | Immunostaining | Western Blot | Co-IP | Molecular Cloning | PCR
Microscopy & Bioimage	Confocal | Airyscan | STED | SIM | STORM | MINFLUX | Expansion Microscopy | Fiji/ImageJ | ZEN Blue | Quantitative Imaging
Clinical & Research	Clinical Trials | TMF/eTMF | Veeva Vault | Research Ethics | Good Clinical Practice | Systematic Review | Scoping Review | Qualitative Research | Scientific Writing
AI & Computing	Python (developing) | Machine Learning (project experience) | Object Detection (project experience) | Bioimage Analysis | Image Quantification | SQL (foundational) | GitHub | Scientific Software Prototyping

PROFESSIONAL EXPERIENCE

Research Admin | Faculty of Medicine, University of Malaya

02/2026 – 04/2026 | Kuala Lumpur, Malaysia

- Synthesized existing literature to identify and validate research instruments for a qualitative study focused on telenursing implementation.
- Co-authored a published scoping review on tele-nursing implementation, benefits, and barriers in postnatal and maternal-child healthcare.

Research Assistant | Hong Kong University of Science & Technology (Guangzhou Campus)

09/2023 – 11/2025 | Guangzhou, China

- Conducted biomedical research involving cell culture, biochemical assays, confocal microscopy, super-resolution microscopy, and quantitative image analysis.
- Investigated protein localization using confocal and super-resolution microscopy.
- Investigated the effects of kinase signalling proteins on microtubule nucleation.
- Contributed to development of automated bioimage quantification using object-detection machine learning.
- Assisted with manuscript review, editing, and preparation of scientific outputs.

Research Assistant | Faculty of Business & Economics, University of Malaya

05/2023 – 07/2023 | Kuala Lumpur, Malaysia

- Assisted with literature collection and survey-question finalization for research on financial initiatives supporting quality caregiving for older adults.
- Participated in participant recruitment and interview processes.
- Contributed to research resulting in publications in WoS-ESCI and Scopus-indexed journals.

Research Assistant | Faculty of Medicine, University of Malaya

12/2022 – 02/2023 | Kuala Lumpur, Malaysia

- Co-authored a systematic review on knee osteoarthritis interventions.

Trial Master File (TMF) Document Specialist 1 | Syneos Health Inc.

04/2021 – 10/2021 | Kuala Lumpur, Malaysia

- Conducted secondary quality control of Trial Master File documents using the Veeva Vault eTMF system.
- Met daily KPIs for document inspection and supported process efficiency.
- Collaborated with project teams to resolve document-quality issues.
- Maintained low/zero weekly error rates during document inspection.
- Received an Accountability Organization Badge.

Genetic Product Specialist Intern | Absolute Genetic Technologies

12/2020 – 02/2021 | Kuala Lumpur, Malaysia

- Interpreted genetic reports in English and Chinese.
- Assisted with survey data collection, analysis, and presentation.
- Created scientific and client-facing content and marketing materials using Canva.
- Collected biological samples to inform product development.

SELECTED SOFTWARE & COMPUTATIONAL PROJECTS

OrganelleScope

Desktop Microscopy Workbench | 2026

• Developed a local desktop microscopy workbench combining the earlier Probabilistic Colocalization Estimator (PCE) and Quick Immunofluorescence Image Line Profiler (QILP).
• Integrated calibrated 8/16-bit multichannel TIFF inspection, ROI measurements, line profiling, colocalization and 3D cuboid/proximity analysis into one workflow.
• Extended the workflow with reviewed image batches, Excel/JSON exports, portable projects and analysis-review reports retaining source, settings and ROI provenance.
• Added an optional local Gemma assistant for human-reviewed interpretation drafts, with source-support and coverage checks kept separate from quantitative measurements.
• Scientific use-case definition and workflow design informed by microscopy experience; implementation developed with AI-assisted coding.

GitHub: github.com/mansonmyr/OrganelleScope

Probabilistic Colocalization Estimator (PCE) | OrganelleScope Predecessor

Scientific Software Prototype | 2026

- Created the earlier ROI-centric tool for multichannel 8/16-bit immunofluorescence TIFF analysis; it subsequently became a predecessor and migration source for OrganelleScope.
- Brought correlation, thresholded overlap and regression-based readouts together with explicit preprocessing and exportable analysis records.
- AI-assisted implementation and prototyping; scientific problem definition and workflow design were driven by microscopy-analysis needs.

GitHub: github.com/mansonmyr/Probabilistic_Colocalization_Estimator

Quick Immunofluorescence Image Line Profiler (QILP) | OrganelleScope Source Tool

Scientific Software Prototype | 2026

- Created a browser-based profiler for 2–4 channel immunofluorescence TIFF images; selected profiling helpers, charting, TIFF export and tests were adapted into OrganelleScope.
- Supports calibrated line ROIs, background subtraction, peak detection, FWHM/AUC measurements, and CSV/JSON result export.
- AI-assisted implementation and prototyping for a microscopy-analysis workflow.

GitHub: github.com/mansonmyr/Quick-IF-Line-Profiler

Automated Cytosolic Microtubule Detection Using Object Detection ML

Biomedical Computer Vision Prototype | 2026

- Prototyped an object-detection workflow for automated identification of cytosolic microtubule structures from fluorescence microscopy images.
- Workflow included microscopy image preparation, manual bounding-box annotation using LabelMe, dataset conversion, model training, model export, and interactive inference demonstration.
- AI-assisted implementation and prototyping; project experience rather than independent ML-engineering expertise.

GitHub: github.com/mansonmyr/Automated-cytosolic-microtubule-using-object-detection-ML

Semi-automated Quantification of In-vivo Microtubule Intensity Using Fiji/ImageJ Macros

Scientific Image Analysis Automation | 2026

- Created Fiji/ImageJ macros for semi-automated quantification of microtubule fluorescence intensity from immunofluorescence z-stack images.
- Workflow uses sum projection, filtering, Find Maxima detection, ROI generation, background correction, centrosomal intensity measurement, and spindle-pole analysis.
- Outputs quantitative measurements including centrosomal intensity, spindle-pole intensity, spindle length, and spindle microtubule intensity.

GitHub: github.com/mansonmyr/Semi-automated-quantification-of-in-vivo-microtubule-intensity-using-Fiji-ImageJ-macros

RESEARCH EXPERIENCE & PROJECTS

iENGAGE Summer Student Research Programme | Wellcome-Wolfson Institute for Experimental Medicine, Queen's University Belfast

07/2023 – 08/2023

- Completed desk-based research on B-cell immune response.
- Received training in data analysis, scientific writing, project design, and experimental techniques.
- Participated in journal clubs, seminars, and a research symposium.

Health Research Ethics Practicum / Summer Research Intern | University of Malaya

07/2022 – 09/2022 | Kuala Lumpur, Malaysia

- Contributed to mHealth research for HIV prevention in partnership with Yale University and the University of Connecticut.
- Conducted transcription and data analysis for focus groups and participated in community advisory board meetings.
- Contributed to a publication in JMIR Formative Research.

Master's Thesis | University of Malaya

2022 | Kuala Lumpur, Malaysia

- Research topic: Ethical Consideration on Clinical Trials in Low-and-Middle Income Countries: an Overview of Systematic Review.
- Presented research at the SIDCER-FERCAP International Conference (2022), and University of Malaya Medical Centre Research Carnival (2023).
- Published in Discover Global Society.

Bachelor's Bioinformatics Project | Sunway University

2020 | Kuala Lumpur, Malaysia

- Investigated G-quadruplex structures in HIV-1 CRF01_AE for potential antiviral development.
- Used QGRS Mapper and NCBI BLASTn to identify potential quadruplex-forming sequences across 1,335 strains.
- Investigated potential G-quadruplex folding across multiple HIV-1 genomic regions.
- Published a narrative review based on the project.

EDUCATION

Master of Artificial Intelligence | Monash University

08/2026 – Present

- Course: Introduction to Python Programming, Introduction to Computer Architecture, Mathematical Foundation for DS/AI, IT Research Method.

Master of Health Research Ethics | University of Malaya

10/2021 – 09/2022 | Kuala Lumpur, Malaysia

- CGPA: 3.61; collaborative degree programme with the Berman Institute of Bioethics, Johns Hopkins University.
- Full scholarship from the Fogarty International Centre of the United States National Institutes of Health (Grant R25TW010891).

Bachelor of Science (Hons) in Medical Biotechnology | Sunway University / Lancaster University, UK

03/2018 – 02/2021 | Malaysia

- Second Class Honours (Division One); CGPA: 3.38.

PUBLICATIONS

- Khati, A., Wickersham, J.A., Rosen, A.O., Luces, J.R.B., Copenhaver, N., Jeri-Wahrhaftig, A., Ab Halim, M.A., Azwa, I., Gautam, K., Ooi, K.H., & Shrestha, R. (2022). Ethical Issues in the Use of Smartphone Apps for HIV Prevention in Malaysia: Focus Group Study With Men Who Have Sex With Men. JMIR Formative Research, 6(12), e42939.
- Ooi, K.H., & Qi, Y. (2023). Unlocking the power of G-quadruplex for anti-HIV1 development. MEDS Clinical Medicine, 4(4), 49-52.
- Che Azmi, A., Ooi, K.H., Ab Aziz, N., Shahimi, S., & Lee, W.L. (2023). Financial Aspects of Elderly Care that Enhances their Quality of Life: A Scientometric Review. Journal of Scientometric Research, 12(3), 691-698.
- Che Azmi, A., Ooi, K.H., Shahimi, S., Ab. Aziz, N., & Silahudin, S. (2024). Financial Challenges in Music and Dance-based Interventions for Older Adults: Systematic Review. Malaysian Journal of Music, 13(1).
- Che Azmi, A., Ooi, K.H., Shahimi, S., & Ab. Aziz, N. (2025). Bibliometric analysis of the financial aspects of quality care for older people. Discover Global Society, 3(117).
- Han, L.J., Ooi, K.H., Krisna, V.B., & Phan, C.W. (2026). Ethical Challenges and Considerations in Decentralized Clinical Trial (DCTs): Insights from a Scoping Review. Therapeutic Innovation & Regulatory Science, 60, 904-924.
- Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). An umbrella review of ethical challenges in clinical trials conducted in low and middle income countries. Discover Global Society, 4(220).
- Ooi, K.H., Marina, J.R., Che, C.C., Nik Farid, N.D., & Abdullah, N.A. (2026). Implementation, Benefits, and Barriers of Tele-nursing in Postnatal and Maternal-Child Health Care: A Scoping Review. BMJ Innovations.
- Lee, J.SY., Gokilavanan, V., Ooi, K.H., & Tan, M.K.M. (2026). Beyond Lockdowns: Equity, Trust, and Social Justice in Malaysia's Pandemic Strategy. In: COVID-19 in the Pacific Rim and Beyond. Collaborative Bioethics, vol. 6. Springer, Cham.
- Lim, W.S., Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). Knowledge, Attitude, Awareness, Barriers, and Willingness to Participate in Clinical Trials: A Quantitative Survey among Undergraduate Students in Malaysia. Malaysian Journal of Medicine and Health Sciences. [Under review]

PROFESSIONAL DEVELOPMENT & AI LEARNING

- GCI World 2026 — Matsuo-Iwasawa Laboratory, The University of Tokyo | Participant | 09/2026 – Present
- Code in Place 2026 — Stanford University | 2026
- Python Basics for Data Science — IBM | Completed
- M1 in SQL — TalentLabs | Completed
- Higher Education Course in Artificial Intelligence — ENEB Business School | Completed
- Teaching English to Foreign Learner Certificate — Foreign Talent Research Centre, MOHRSS China | Completed
- Clinical Research Associate Certificate — CCRPS | Completed
- Social and Behavioral Responsible Conduct of Research — CITI Program | Completed
- Result-based Project Management — Inter-American Development Bank | Completed

INTERNATIONAL EDUCATION & LANGUAGE DEVELOPMENT

Hanyang International Summer School | Hanyang University

06/2026 – 07/2026 | Seoul, South Korea

- Completed beginner Korean language course.

TECHNICAL SKILLS

Laboratory: Cell Culture | Immunostaining | Crosslinking Co-IP | Western Blot | Sucrose Gradient Centrifugation | Molecular Cloning | PCR
Microscopy: LSM980 Confocal & Airyscan (Zeiss) | Stellaris 8 Confocal (Leica) | Axio Observer 7 | STED (Abberior STEDyCON) | SIM (Zeiss Elyra 7) | STORM (Abbelight) | MINFLUX (Abberior) | Expansion Microscopy
Flow Cytometry: BD FACSymphony A1 | FlowJo | Cell-cycle analysis
Bioimage Analysis: Fiji/ImageJ | ZEN Blue
Bioinformatics: GO & KEGG analysis | Clustal MSA | NCBI BLASTn | QGRS Mapper | TCGA
AI / Data: Python (developing) | Machine Learning (project experience) | Object Detection (project experience) | Automated Image Quantification | SQL (foundational) | GraphPad Prism | SPSS
Administrative / Digital: Veeva Vault | Tableau | Flourish | Canva | Microsoft Office | G Suite | Lucidchart | MindNode | Google Sites | Gantt Chart

LANGUAGES

- English — IELTS 7.0 (2026)
- Mandarin
- Malay
- Cantonese
- Korean — Beginner; Hanyang International Summer School (06/2026 – 07/2026)

CONFERENCE & SERVICE

Conference Support Volunteer | International Conference on Cell Organization & Genome Integrity 2024, HKUST(GZ)

12/2024 | Guangzhou, China

- Facilitated registration, session logistics, attendee support, and daily conference operations.
- Provided bilingual Chinese-English assistance to international delegates.
- Managed attendee flow to support smooth session transitions.

Oral Presenter | 22nd WHO SIDCER-FERCAP 2022, K-MEDI Hub

11/2022 | Daegu, Korea

- Presented topic on ethical consideration in clinical trial among low- and middle-income countries.